

MANIPAL UNIVERSITY JAIPUR
Department Electronics & Communication Engineering
Course Hand-out

A. Basic Details:

Program Name:	B.Tech. (ECE)
Course Name:	Computer Architecture & Processor
Course Code:	ECE2108
LTPC (Lecture Tutorial Practical Credits):	3 0 0 3
Session:	July-Dec 2027
Class:	3 rd Sem (2 nd Yr)
Course Coordinator:	Dr Rohit Mathur
Course Instructor(s):	Dr Manish Tiwari & Dr Rohit Mathur
Additional Practitioner(s):	Sessions from <i>Industry Expert</i> from eInfochip / SoCTEAMUP / TrueChip / VLSI System Design, etc.

B. Introduction:

This course on Computer Architecture & Processor is about the structure and function of computers. Its purpose is to present, the nature and characteristics of modern-day computer systems.

C. Course Outcomes: By the end of the course, students will be able to

<i>CO Statement</i>	<i>CO</i>	<i>Level</i>	<i>Target Attainment %</i>	<i>Target Attainment level</i>
Demonstrate understanding of Register Transfer and Microoperations.	1	L2	75	2
Demonstrate understanding of control unit and its subsystem design.	2	L2	75	2
Explain concepts and impact of cache memory, pipelining and parallel processing on system performance.	3	L2	75	2
Explain the operation of various addressing modes of 8086 microprocessor.	4	L2	85	3
Develop Logical and Conditional assembly language programming skills for 8086 microprocessor.	5	L3	75	2

Information about attainment levels:

Attainment (%)	< 60 %	≥ 60% < 70%	≥ 70% < 80%	≥ 80
Level	0	1	2	3

D. Program Outcomes and Program Specific Outcomes:

- [PO. 1]: Engineering knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
- [PO. 2]: Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
- [PO. 3]: Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
- [PO. 4]: Conduct investigations of complex problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

- [PO. 5]: **Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
- [PO. 6]: **The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.(WK1, WK5, and WK7).
- [PO. 7]: **Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
- [PO. 8]: **Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.
- [PO. 9]: **Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.
- [PO. 10]: **Project management and finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to multidisciplinary environments.
- [PO. 11]: **Life-long learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change.(WK8).

PROGRAM SPECIFIC OUTCOMES: B.Tech. (Electronics and Communication Engineering)

- [PSO. 1]: An ability to understand the concepts of basic Electronics & Communication Engineering and to apply them to various areas like Signal processing, VLSI, Embedded systems, Communication Systems, Digital & Analog Devices, etc.
- [PSO. 2]: An ability to solve complex Electronics and Communication Engineering problems, using latest hardware and software tools, along with analytical skills to arrive cost effective and appropriate solutions.
- [PSO. 3]: Wisdom of social and environmental awareness along with ethical responsibility to have a successful career and to sustain passion and zeal for real-world applications using optimal resources as an Entrepreneur.

E. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): (Quiz + Assignments) / (Research + Presentation)	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. Syllabus:

Computer Architecture: Introduction, computer types, functional units, Von Neumann and Harvard architectures, RISC and CISC Architectures. Register Transfer and Microoperations, Design of Basic Computer, Design of Accumulator Logic, Control unit design, Microprogrammed Control, Parallel Processing, Pipelining, Input-Output Organization, Memory Organization. **8086 Microprocessor:** 8086 Architecture, 8086 Instruction Set: types of instructions and addressing modes, assembler and assembler directives, assembly language programming. **RISC V:** Introduction to the Reduced Instruction Set Computer, RISC V architecture and features.

References:

1. M. M. Mano, "Computer System Architecture", Pearson, (3e), 2007.
2. K. M. Bhurchandi & A. K. Ray, "Advanced Microprocessors and Peripheral Devices", McGraw Hill, (3e), 2018.
3. D. A. Patterson & J. L. Hennessy, "Computer Organization and Design: The Hardware/Software Interface- RISC-V Edition", Morgan Kaufmann, (1e), 2018.
4. V.C. Hamacher, Z. Vranesic & S. Zaky, "Computer Organization", McGraw Hill, (5e), 2002.
5. J. P. Hayes, "Computer Architecture and Organization", TMH, (3e), 1998.

G. Lecture Plan:

L. No.	Topic	Session Outcome	Mode of delivery	Corr. CO	Mode of assessing CO
1.	Introduction to Computer Architecture, Computer Types, Functional Units.	Explain the general functions and structure of a digital computer.	Lecture based teaching-learning	CO1	Quiz, MTE, ETE
2.	Von Neumann and Harvard architectures, RISC and CISC Architectures.	Identify basic components of a computer & explain the purpose of different functional units.	Lecture based teaching-learning	CO1	Quiz, MTE, ETE
3.	Register Transfer Language and Register Transfer.	Register Transfer Language and Register Transfer.	Lecture based teaching-learning	CO1	Assignment, MTE, ETE
4.	Bus and Memory Transfers	Design a basic Bus and Memory Transfers circuit	Lecture based teaching-learning	CO1	Assignment, MTE, ETE
5.	Arithmetic Microoperations	Design a basic Arithmetic Microoperations circuit	Learning through problem-solving	CO1	Assignment, MTE, ETE
6.	Logic Microoperations	Design a basic Logic Microoperations circuit	Inquiry based learning	CO1	Assignment, MTE, ETE
7.	Shift Microoperations	Design a basic Shift Microoperations circuit	Learning through problem-solving	CO1	Assignment, MTE, ETE
8.	Arithmetic Logic Shift Unit	Design a basic Arithmetic Logic Shift Unit circuit	Inquiry based learning	CO1	Assignment, MTE, ETE
9.	Basic Computer Organization	Explain the concept of Computer Organization	Inquiry based learning	CO2	Quiz, MTE, ETE
10.	Design of Basic Computer	Explain the role and responsibility of different components of a Basic Computer and their union.	Lecture based teaching-learning	CO2	Quiz, MTE, ETE
11.	Design of Accumulator Logic	Explain the Design concepts of a basic Accumulator Logic	Lecture based teaching-learning	CO2	Quiz, MTE, ETE
12.	Microprogrammed Control	Explain the concept & need of Microprogrammed Control	Lecture based teaching-learning	CO2	Assignment, MTE, ETE
13.	Control Memory	Explain the concept & need of Control Memory	Lecture based teaching-learning	CO2	Assignment, MTE, ETE
14.	Control unit design	Explain the nuances of Control unit design	Lecture based teaching-learning	CO2	Assignment, MTE, ETE
15.	Program Control	Explain the concept of Program Control	Lecture based teaching-learning	CO2	MTE, ETE
16.	* Parallel Processing	Explain the concept of Parallel Processing	Lecture based teaching-learning	CO3	Quiz, ETE
17.	* Pipelining	Explain the role of Pipelining in speedup	Lecture based teaching-learning	CO3	Assignment, ETE
18.	Instruction Pipeline	Explain the concept of Instruction Pipeline	Inquiry based learning	CO3	Quiz, ETE
19.	Pipeline Hazards	Differentiate different types of Pipeline Hazards	Inquiry based learning	CO3	Quiz, ETE
Mid Term Examination					
20.	Input-Output Organization: Peripheral Devices	Explain the need of Input-Output Organization	Lecture based teaching-learning	CO3	Quiz, ETE
21.	Asynchronous & Synchronous Data Transfer	Explain the need of Asynchronous & Synchronous Data Transfer	Lecture based teaching-learning	CO3	Assignment, ETE
22.	Modes of Transfer	Explain concept & need of different Modes of Transfer	Lecture based teaching-learning	CO3	Assignment, ETE
23.	Memory Organization: Memory Hierarchy	Explain concept & need of Memory Organization	Inquiry based learning	CO3	Quiz, ETE

24.	Main Memory & Auxiliary Memory	Explain concept & need of Main Memory & Auxiliary Memory	Inquiry based learning	CO3	Quiz, ETE
25.	Associative Memory	Explain concept & need of Associative Memory	Lecture based teaching-learning	CO3	Quiz, ETE
26.	* Cache Memory	Explain how speedup is achieved via Cache Memory	Lecture based teaching-learning	CO3	Quiz, ETE
27.	8086 Microprocessor: Internal Architectural Model, operational details,	Demonstrate knowledge of architecture of 8086 & it's working principle.	Flipped Classroom	CO 4	Assignment, ETE
28.	8086 Addressing modes - 1	Describe the various types of addressing modes common of 8086.	Lecture based teaching-learning	CO 4	Assignment, ETE
29.	8086 Addressing modes - 2	Describe the various types of addressing modes common of 8086.	Learning through problem-solving	CO 5	Assignment, MTE, ETE
30.	Machine Language, Assembly Language and Assembler	Describe the various Assembly Language constructs.	Lecture based teaching-learning	CO 5	Assignment, MTE, ETE
31.	Programming Arithmetic and Logic Operations	Develop basic Programming skills	Technology based learning	CO 5	Assignment, MTE, ETE
32.	Program Loops & Subroutines	Develop basic Programming skills	Technology based learning	CO 5	Assignment, MTE, ETE
33.	Instruction Set: types of instructions & assembler directives.	Develop basic 8086 Programming skills.	Lecture based teaching-learning	CO 5	Assignment, ETE
34.	Assembly Language Programming	Develop intermediate 8086 Programming skills.	Learning through problem-solving	CO 5	Assignment, ETE
35.	RISC-V: Introduction to the Reduced Instruction Set Computer.	Demonstrate knowledge on need & USP of RISC-V architecture.	Lecture based teaching-learning	CO 5	Quiz, ETE
36.	* RISC-V architecture and features.	Demonstrate knowledge of architecture of RISC-V & it's working principle.	Lecture based teaching-learning	CO 5	Quiz, ETE
End Term Examination					

Note: lectures marked as * indicates that the lectures proposed to be covered by additional practitioners (Industry Fellow/ Visiting Faculty/ Adjunct Faculty, etc.)

H. Course Articulation Matrix: Course articulation indicates the mapping of CO statement(s) to some PO/ PSO statement(s).

CO	Statement	Correlation with Program Outcomes & Program Specific Outcomes														
		P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P S O 1	P S O 2	P S O 3	
1	Demonstrate understanding of Register Transfer and Microoperations.	3	2	2	1	-	-	-	-	-	-	-	1	1	-	
2	Demonstrate understanding of control unit and its subsystem design.	3	2	2	1	-	-	-	-	-	-	-	1	1	-	
3	Explain concepts and impact of cache memory, pipelining and parallel processing on system performance.	3	2	2	2	-	-	-	-	-	-	-	1	1	-	
4	Explain the operation of various addressing modes of 8086 microprocessor.	3	2	2	1	-	-	-	-	-	-	-	1	1	-	
5	Develop Logical and Conditional assembly language programming skills for 8086 microprocessor.	3	2	2	2	1	-	-	-	-	-	-	1	1	-	

Here value 0 indicates no correlation, 1 as low correlation, 2 as moderate correlation, and 3 as substantial / strong correlation.